Md Shamimul Islam

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Research Interests

Cybersecurity & Intrusion Detecsion System (IDS) | LLM & RAG | ML | | Biometrics

Education

2025 – Present
- **MS in Computer Science, Florida Polytechnic University**
  CGPA: 3.77 out of 4.00

2020 – 2022
- **Masters in Applied Statistics and Data Science, Jahangirnagar University (JU)**
  CGPA: 3.96 out of 4.00
  Thesis title: *Video-based Bangla Sign Language Recognition using Deep learning Algorithms.*

2015 – 2020
- **B.Sc. Computer Science, Manarat International University (MIU).**
  CGPA: 3.99 out of 4.00
  Thesis title: *Pose-based Multi-level Features Fusion For Exam Cheat Recognition Using CNN.*

Research Publications

Peer-Reviewed Publications (Total Citation: 97 | h-index: 4 | i10-index: 2)

1. **M. S. Islam**, A. J. M. A. Joha, N. Hossain, M. M. Hasan, and S. Abdullah, "Multi-view bangla sign language recognition: A new word-level video dataset," in *2024 IEEE The 6th International Conference on Activity and Behavior Computing*, Nakatsu and Kitakyushu, Kyushu, Japan, 2024. ↗DOI: 10.1109/ABC61795.2024.10652018.

2. M. M. Hasan, **M. S. Islam**, and S. Abdullah, "Robust pose-based human fall detection using recurrent neural network," in *IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things*, Dhaka, Bangladesh, 2019. ↗DOI: 10.1109/RAAICON48939.2019.23.

3. **M. S. Islam**, M. M. Hasan, and S. Abdullah, "A deep spatio-temporal network for vision-based sexual harassment detection," in *IEEE International Conference on Emerging Technology in Computing, Communication and Electronics*, 2021. ↗DOI: 10.1109/ETCCE54784.2021.9689891.

4. **M. S. Islam** and I. Elwarfalli, "Deep learning-powered ecg-based biometric authentication," in *IEEE International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM)*, DUET, Dhaka, Bangladesh, 2023. ↗DOI: 10.1109/NCIM59001.2023.10212845.

5. N. H. M. Arafat, S. Ileas, Mujazid, and **M. S. Islam**, "Blosecr: A conceptual framework of identity documents' security on blockchain in refugee issues and conserving fundamental rights," in *The International Conference for Intelligent Technologies*, Karnataka, India, 2022. ↗DOI: 10.1109/CONIT55038.2022.9847731.

Under Review / In Preparation

1. **M. S. Islam**, L. G. Jaimes, and A. S. Dina, "MA-IDS: Multi-agent RAG framework for IoT network intrusion detection with an experience library," Submitted to 2026 IEEE International Conference on Systems, Man, and Cybernetics (SMC), April 2026. Bellevue, WA, USA, 2026.
2. **M. S. Islam**, A. H. Sohag, and Z. Subah, "Permissionguard: Fine-tuned LLMs and BERT for android permission-based malware detection," Manuscript in preparation, 2025.

Datasets

1. A. J. M. A. Joha, M. N. Hossain, **M. S. Islam**, S. Abdullah, and M. M. Hasan, *Multi-view bangla sign language (mv-bsl) dataset*, Public Dataset, Dataset available at `https://github.com/Mahedi-61/MV-BSDL`, 2024.
2. S. Abdullah, **M. S. Islam**, and M. M. Hasan, *Bangladeshi vehicle license plate dataset*, Public Dataset, Dataset available at `https://github.com/Mahedi-61/Bangla_License_Plate_Dataset`, 2024.
3. **M. S. Islam** and M. M. Hasan, *Sexual harassment video dataset (shvd)*, Public Dataset, Dataset available at `https://github.com/Shamim-38/Sexual-Harassment-Video-Dataset-SHVD-`, 2024.

Employment History

2025 – · · · ·	• **GA,** Department of Computer Science, Florida Polytechnic University.
2024 – 2025	• **Lecturer,** Department of Computer Science and Engineering (CSE), Manarat International University (MIU).
2023 – 2024	• **Lecturer,** Department of CSE, Asian University of Bangladesh (AUB) Convener, AUB Programming Club.
2022 – 2022	• **Adjunct Faculty,** Department of Islami Studies (DIS), MIU
2020 – 2022	• **Adjunct Faculty,** Department of CSE, MIU.

Research Projects

2025 – Present	• **MA-IDS: Multi-Agent RAG Framework for IoT Intrusion Detection (Thesis):** Developed a multi-agent IDS integrating GPT-4o with a FAISS-based Retrieval-Augmented Generation (RAG) library. Features a Traffic Classification Agent and an Error Analysis Agent for dynamic rule generation. Achieved Macro F1-Scores of 89.75% (NF-BoT-IoT) and 85.22% (NF-ToN-IoT), outperforming zero-shot baselines by up to 80%. Submitted to IEEE SMC 2026. Advisors: Dr. Ayesha S. Dina, Dr. Luis G. Jaimes (Florida Poly).
2026 – Present	• **PermissionGuard: Fine-Tuned LLMs & BERT for Android Malware Detection:** Compared LoRA fine-tuning of LLaMA 3.1-8B and full fine-tuning of BERT-base against classical ML baselines for Android permission malware detection. LoRA achieved 97.00% (NATICUSdroid) and 88.39% (Mahindru) accuracy, exceeding baselines by 19+%. Course project; manuscript in preparation.

Research Projects (continued)

2024 – 2025 • **AI Career Counselor & Guidance Platform:** Built a LLaMA-powered platform providing personalized IT career roadmaps and labor market insights. Trained on an 8,000-point dataset curated from leading educational and job platforms.

2024 • **Detection of Phishing URLs for Secure Browsing:** Developed an ML system to detect fraudulent URLs using Logistic Regression, Random Forest, SVMs, and Neural Networks with advanced feature selection to optimize classification accuracy.

2022 – 2023 • **Real-time Bangla Sign Language Translation System:** Proposed a novel Transformer architecture to translate pose-based Bangla sign language video descriptors into sentences. Won the "Artificial Intelligence for Bangla Competition 2.0" (Short version).

2022 • **Bangla Social Network Post Analysis using Deep Learning:** Curated a custom dataset and trained a Transformer-based language model to classify political sentiment (pro/anti-government) and toxicity levels in real-time social media posts.

2018 – 2019 • **Automatic License Plate Recognition (ALPR):** Developed and deployed a real-time ALPR system trained on a Bangladeshi license plate dataset. Published and presented at an IEEE conference (2019).

Miscellaneous Experience

Awards and Achievements

2023 • **Winner of AI for Bangla 2.0 Competition:** As team leader, we secured the position in winner list in the national-level AI For Bangla 2.0 competition in 2023, which was directed by the Bangladesh Government, leading a team in developing an innovative AI solution for the Bangla sign language. Competition involved more than a hundred teams.

2020 • **Islamic Development Bank (IsDB) Masters Scholarship:** Received the highly competitive IsDB Scholarship in 2020, awarded to outstanding students pursuing graduate studies in Master of Science. Scholarship valued at $6,000.

2018 • **University Merit Scholarship:** Due to placement in the top 1% of students in the institute, tuition fees were waived in undergraduate studies.

Certification

2015 • **Champion** in Great Victory Day Essay Competition. Awarded by Manarat International University.

Co/Extracurricular Activities

2020 • **Elected President** at MIU CSE Club, Manarat International University (MIU), Ashulia, leading department's students in organizing events and activities to promote interest in the field of computer science and and organized some successful events, such as MIU CSE Fest 2020, Intra university programming contest and so on.

2017 • **Co-founder** at Hasimukh (Smile), a social welfare organization established in Rangpur, Bangladesh, working to support the underprivileged people in Bangladesh.

2018 • **Participant** at The ICPC Bangladesh IUBAT NCPC National Programming Contest, Dhaka, Bangladesh.

2019 • **Participant** at Hackathon, organized by Google Developer Group (GDG) Sonargaon, Dhaka, Bangladesh.

Miscellaneous Experience (continued)

- **Organizing Member** at Code Camp organized by MIU CSE Club, Manarat International University, Ashulia.

Talks and Poster Presentations

2026
- **Invited Speaker** at 2026 Engineering and Science Student Research Symposium, Florida Polytechnic University, Lakeland, FL. Talk: "Adaptive Reasoning for Network Traffic Security: A Multi-Agent RAG Architecture with Persistent Knowledge Induction." Advisor: Dr. Ayesha S. Dina.
- **Poster Presentation:** "MA-IDS: Multi-Agent RAG Framework for IoT Network Intrusion Detection with an Experience Library." Florida Polytechnic University, Lakeland, FL.

2022
- **Invited Speaker** at Webinar on A Publication Road Map for Scientific Papers, organized by MIU CSE club, MIU, Ashulia.

Skills

Languages	• Strong reading, writing and speaking competencies for Bengali and English.
Coding	• Python, Java, PHP, C, R, SQL, XML/XSL.
System Administration	• Unix/Linux, Windows
Document Processing	• LaTeX, MS Office
Developer Tools	• Git, Google Cloud Platform
Libraries	• NumPy, OpenCV, Pandas, Matplotlib
AI Framework	• PyTorch, TensorFlow
Object Detection Algorithms	• YOLO, RetinaNet, Mask R-CNN
Pose Detection Algorithm	• OpenPose, MediaPipe
Misc.	• Academic research, teaching, training, consultation, LaTeX typesetting and publishing.

References

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